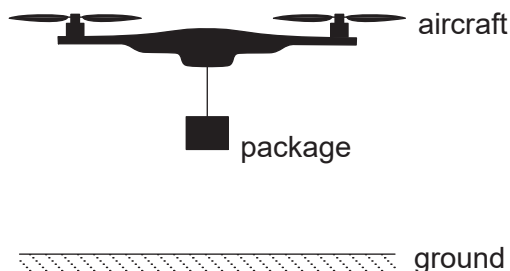


17. A company delivers packages to customers using a small unmanned aircraft. Rotating horizontal blades exert a force on the surrounding air. The air above the aircraft is initially stationary.



The air is propelled vertically downwards with speed v . The aircraft hovers motionless above the ground. A package is suspended from the aircraft on a string. The mass of the aircraft is 0.95 kg and the combined mass of the package and string is 0.45 kg. The mass of air pushed downwards by the blades in one second is 1.7 kg.

- (a) (i) State the value of the resultant force on the aircraft when hovering. [1]

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- (ii) Outline, by reference to Newton's third law, how the upward lift force on the aircraft is achieved. [2]

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- (iii) Determine v . State your answer to an appropriate number of significant figures. [3]

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(This question continues on the following page)

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(Question 17 continued)

(iv) Calculate the power transferred to the air by the aircraft.

[2]

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(b) The package and string are now released and fall to the ground. The lift force on the aircraft remains unchanged. Calculate the initial acceleration of the aircraft.

[2]

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